# Aaron Advait Anish

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 6.66 |
| Narayana Junior College | Class XII (Telangana State Board of Intermediate Education (TSBIE)) | 2024 | 95.70% |
| FIITJEE World School | Class X (Board of Secondary Education, Telangana) | 2022 | 9.80% |

## POSITIONS OF RESPONSIBILITY

**Consulting Coordinator | Finance and Consulting Club | IITH**

- Co-Authored the CaseBook used in preparation for consulting and product management interviews
- Conducted CaseKnights: A campus-wide initiative to educate students with regards to approaching Product Management interviews
- Conducted PRODX '26: A nationwide case competition with 4000+ applications, including from various IITs and IIMs.
- Assisted in the successful organization of HORIZON, the flagship event of FCC for the years 2025 and 2026.

**Sponsorship Coordinator | Elan&nVision**

- Assisted in the organisation of Elan&nVision '26, the flagship techno-cultural fest of IIT Hyderabad.
- Assisted in the organisation of nVisionX '26, the flagship techno fest of IIT Hyderabad.

## SKILLS

**Languages:** C | C++ | Java | Python | RISC-V | SQL | Flex

**Developer Tools:** PostgreSQL | Docker, Git | VS Code

## RELEVANT COURSES

**Software Core:** Software Development Fundamentals, Data Structures, Algorithms, Database Management Systems.

**Systems Core:** Operating Systems, Computer Architecture, Compilers.

**Mathematics Core:** Discrete Mathematics for comp. sci., Probability & Random Variables, Linear Algebra, Theory of Computation.

## EXPERIENCE

**Business Consulting Intern | Factoryal Business Incubator Pvt. Ltd.**

- Applied **first-principles thinking** and analytical problem-solving to evaluate organizational structures and optimize business processes.
- Researched **Sales** and **HR** functions, mapping workflows and designing **AI-powered micro-process automations** using **n8n**.
- Developed client outreach strategies and contributed to sales pitch decks for presenting workflow automation solutions.
- Collaborated with the founding team on business operations, process improvement, and automation initiatives.

## PROJECTS

**Query Optimization & Performance Analysis** | PostgreSQL, SQL, Python | Course Project

- Analyzed **query execution performance** on TPC-H workloads using **EXPLAIN ANALYZE**, comparing sequential scans, index scans, clustered indexes, and varying query selectivities.
- Developed **Python automation** to generate parameterized queries, execute benchmarks across selectivity levels, and extract **PostgreSQL execution plans** for performance analysis.
- Implemented **query plan comparison and plan-switch detection**, identifying points where the PostgreSQL optimizer changed execution strategies as query parameters varied.
- Evaluated optimizer decisions using **actual execution times**, classifying plan switches as **EARLY, DELAYED, or CORRECT** and determining optimal switch points.

**C++ Poker Win Probability Bot** | C++ | Personal Project

- Engineered a core C++ poker bot to calculate real-time win probabilities by running 10,000-iteration **Monte Carlo simulations** against simulated opponents.
- Implemented the highly optimized **Cactus Kev** evaluation algorithm, utilizing advanced **bitwise manipulations** to encode card ranks, suits, and prime mappings into 32-bit integers.
- Achieved O(1) time complexity for hand ranking resolution by leveraging prime number multiplication logic and direct lookup **table calls** instead of nested conditional loops.
- Developed custom deck-shuffling and array management algorithms to efficiently evaluate all 21 possible 5-card combinations from a 7-card pool.

**ResearchPilot |** End-to-End Hybrid RAG & Vector Search Platform | Personal Project

- Architected a low-latency **Retrieval-Augmented Generation (RAG)** system in **FastAPI** and **Next.js** using **embedded Qdrant** for semantic document querying.
- Implemented Hybrid Retrieval combining dense vector embeddings (text-embedding-3-small) with sparse keyword matching **(BM25)**, fused via **Reciprocal Rank Fusion (RRF)**.
- Applied a cross-encoder reranking layer (bge-reranker-large) to optimize context relevance, reducing hallucination rate and boosting Precision@K on technical PDFs.
- Designed citation-grounded prompt pipelines with deterministic page-level metadata tracking to guarantee complete response auditability.

## EXTRACURRICULAR

- **Classical Piano, Trinity College of Music London:** Grade-6
- **Black Belt, Teakwondo:** 2 time Guiness World Record Holder.